

* Spelling Error Detection and Correction:-

In Computer based Information Systems, errors of typing and Spelling constitute a Very Common Source of Variation between the strings.

- i) Substitution of a Single letter :- Concept : Concpete
- ii) Omission of a Single letter. :- Error : Error
- iii) Insertion of a Single letter :- Errorn
- iv) transposition of 2 adjacent letters :- are : aer

Similarly we have Spelling errors:

2 types of Spelling errors:

① Non-word Error :- When a word does not appear in Lexical / Lexicon (or) not Valid in orthographic word. it is termed as non-word.

② A Real word Error : Results in actual words of the language such as piece for peace and meat for meet. Real-word errors may cause local Syntactic errors, global Syntactic error it becomes impossible to decide the word is wrong without some Contextual Information.

Spelling correction consists of detecting and correcting errors. Error detection is the process of finding misspelled words and error detection is the process of suggesting correct words to a misspelled one.

2 types :-

① Isolated Error detection and correction :- Each word is checked separately, independent of context.

a) The strategy is maintaining larger lexicon containing all words. which would take long time to compile and occupy more space.

b) The larger the Lexicon, the more likely it is that error goes undetected. because chance of word being found is greater.

c) The Realword errors Such as Ex:- These written as theses.

② Context dependent Error detection :- Utilizes context of a word to detect and correct errors. This requires grammatical analysis and more complex and language dependent.

Different types :-

- ① Minimum edit distance
- ② Similarity key technique
- ③ n-gram based technique
- ④ Neural Nets
- ⑤ Rule based technique.

Minimum Edit Distance :-

The minimum edit distance is the no of Insertions, deletions and Substitutions. required to change one string into another.

→ Inserting character
(Horizontal arrow)

(Vertical arrow)

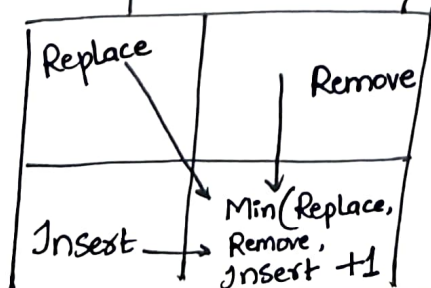
↓ Removing character.

Ex:- We have to convert **a d c e g** into **a b c f g**
Let us draw a memorization matrix.

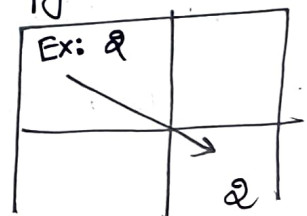
Step ¹	NULL	a	b	c	f	g
NULL	0	1	2	3	4	5
a	1	0				
d	2					
c	3					
e	4					
g	5					

2 operations are required to convert string
a d c e g
a b c f g

Rule 1 :- if row element $\neq$ column element



Rule 2 :- if $r = c$
Just copy the Diagonal element



Step 1:- From NULL string to get a, we have to do 1 operation
 $\rightarrow a$ and so on to get g $\rightarrow$ 5 operation

Step 2:- In the same way, to remove a we have to do 1 operation $\downarrow 1$ and so on to get $\downarrow 5$ to get g.

Step 3:- Cross check with the element. from row and column
if both are equal copy the diagonal element. else.
Copy the minimum value from the 3 cells and add one to it.

Finally check the string

① check the elements diagonally. for backtracking.

② check the last element

$$g = g$$

③ $e \neq f$ so replace it.

④ $c = c$

⑤ $d \neq b$ so replace it

⑥ $a = a$

	NULL	a	b	c	f	g
NULL	0					
a		0				
d			1			
c				1		
e					2	
g						2

as per the last element 2 operations are required.